

Equation sheet

Kinematics

For constant acceleration, a_x : $v_x = v_{0x} + a_x t$, $\Delta x = \frac{1}{2}(v_{0x} + v_x)t$, $\Delta x = v_{0x}t + \frac{1}{2}a_x t^2$, $v_x^2 = v_{0x}^2 + 2a_x \Delta x$
(same for y-motion)

Quadratic equation: $ax^2 + bx + c = 0 \rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Relative velocity: $\vec{v}_{AB} = \vec{v}_{AE} + \vec{v}_{EB}$

Dynamics

Newton's 2nd law: $\sum \vec{F} = m\vec{a}$

Friction: $f_k = \mu_k n$ (kinetic), $f_s \leq \mu_s n$ (static) (n or N is the normal force)

For circular motion: $a_c = \frac{v^2}{r}$

Spring Force: $F_s = -kx$

Work-Energy

Work: $W = \vec{F} \cdot \vec{d} = (F \cos \theta)d$, $W_{nc} = E_f - E_i$

Mechanical Energy E: $KE = \frac{1}{2}mv^2$ (kinetic energy), $PE_g = mgy$ (gravitational potential energy)

$PE_s = \frac{1}{2}kx^2$ (spring potential energy)

(Average) Power: $\bar{P} = \frac{W}{\Delta t}$, $\bar{P} = F\bar{v}$

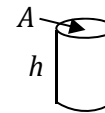
Momentum

Momentum: $\vec{p} = m\vec{v}$,

Impulse: $\vec{J} = \vec{F}\Delta t = \vec{p}_f - \vec{p}_o$

DATA

$V_{sphere} = \frac{4}{3}\pi r^3$, $A_{sphere} = 4\pi r^2$, $A_{circle} = \pi r^2$, $V_{cylinder} = Ah$
circumference = $2\pi r$



$g = 9.80 \text{ m/s}^2$